

Modulation of BP Acrylic PSA Modulus for Enhanced Drop Impact Strength of Flexible OLED Panels

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Abstract

Flexible OLED represents an emerging development in the field of flat panel displays (FPD), offering numerous advantages over LTPS LCD such as lighter weight, thinner profile, greater flexibility, and usability in multiple forms including bending and folding. However, the trend toward increasingly thinner and lighter product designs has posed severe challenges to mechanical strength, particularly the impact resistance of the panel. In this paper, the JKL contact theory is cited to illustrate the reduction of modulus, the improvement of the wettability of PSA adhesive, and the better adhesion between the PSA and the panel during attachment.. So the stress transmission from the impact of the falling ball to the attachment interface is more uniform. Secondly, the John Ferry high-frequency modulus dissipation theorem is introduced, and the time temperature equivalence theorem is referenced. The modulus of PSA adhesive is ensured to be within 1000Kpa below -20 °C, which has good energy dissipation and buffering effect, reduces the impact energy transmitted to the panel, and can improve the impact strength of panel falling balls by more than 20%.

Author Keywords

flexible OLED ; impact resistance; PSA;energy dissipation

1. Introduction

The BP support film is composed of PET material and acrylic pressure-sensitive adhesive. Figure 1 illustrates the mainstream module stack structure of current flexible OLED panels, wherein the acrylic adhesive within the substrate film serves as the direct bonding contact layer with the OLED display panel. The stress-strain it experiences, along with any undissipated energy, can be directly transmitted to the internal film layers of the panel, leading to crack formation and consequently causing display defects such as bright spots and dark spots. However, there is currently a lack of systematic theoretical explanation or practical guidance within the industry regarding how variations in the relevant properties of this acrylic UV PSA affect the impact strength of flexible OLED screens.

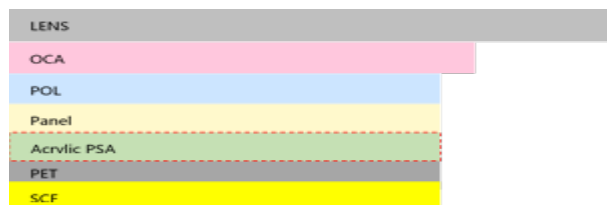


Figure 1. Flexible OLED Module Stack

Herein, we have focused our research on enhancing the impact resistance of OLED panels through acrylic pressure-sensitive adhesive (PSA), with findings indicating that modulus adjustment constitutes a critical factor in improving impact performance.

Serving as the direct contact layer with the OLED panel, the bonding effectiveness of the acrylic PSA emerges as a pivotal consideration. In accordance with the JKR contact theory—which characterizes the contact behavior between two elastic spheres under the influence of surface energy—we present the formula for contact radius "a". This formulation explicitly incorporates the material's modulus "E".

$$a^3 = \frac{R}{K} (F + 3\pi\gamma R + \sqrt{6\pi\gamma R F + (3\pi\gamma R)^2}), K = \frac{4E}{3(1-\nu^2)} \quad (1)$$

Based on the JKR contact theory, where γ represents the surface energy, R is the radius, and F denotes the applied load, the contact radius 'a' increases as the modulus E decreases under constant surface energy and load conditions. This implies a larger contact area and improved wetting coverage. The same principle applies to the Acrylic PSA adhesive. When the adhesive contacts the panel, a lower modulus enhances its fluidity, allowing it to effectively wet and fill the micro-scale irregularities on the panel surface. As illustrated in Figure 2, the actual contact area exceeds the nominal surface area. In contrast, when the modulus is higher (as shown in Figure 3), the adhesive exhibits poor fluidity during lamination, resulting in inadequate filling and an actual contact area smaller than the nominal surface area. This condition carries a risk of stress concentration.

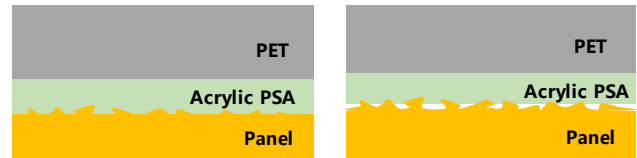


Figure 2. High-Conformability **Figure 3.** Low-Conformability

Table1.Wettability and Its Interfacial Effects

Condition	High Wettability	Low Wettability
Contact area	Large area	Limited Area
Stress	Low	High
Energy Dissipation	High	Low
Impact Resistance	High	Low

Furthermore, as a viscoelastic material, PSA exhibits both viscous and elastic properties. The viscous component enables the adhesive to flow in a liquid-like manner, where internal molecular chains move relative to each other at very high speeds, disentangling and generating friction. The intense motion of these

molecular chains must overcome significant internal frictional resistance, thereby converting macroscopic, ordered kinetic energy into microscopic, disordered thermal energy (2). Integrating this mechanism into the drop impact model of the panel: if the kinetic energy of the impacting ball is effectively dissipated, the amount of energy transmitted to the panel is reduced. As a result, the panel demonstrates higher performance in drop impact tests. This energy dissipation behavior can be further explained by referring to the John Ferry high-frequency energy dissipation theorem.

$$W_d = \pi \gamma_0^2 G'' \quad (\gamma_0 \text{ is the strain amplitude}) \quad (3)$$

The above formula shows, a higher value of G'' indicates a greater ability of the material to convert mechanical energy into heat per cycle. However, the dissipated energy is extremely sensitive to the deformation amplitude γ_0^2 (strain = compression displacement / sample thickness). If the strain amplitude doubles, the dissipated energy increases by a factor of four, demonstrating that deformation is the key driver of energy dissipation. If the strain amplitude of the pressure-sensitive adhesive is enhanced, meaning its modulus is reduced (Figure 4). The aforementioned theory has been validated through physical prototype fabrication and drop impact testing, confirming that a reduction in modulus results in superior impact resistance levels for OLED panels.

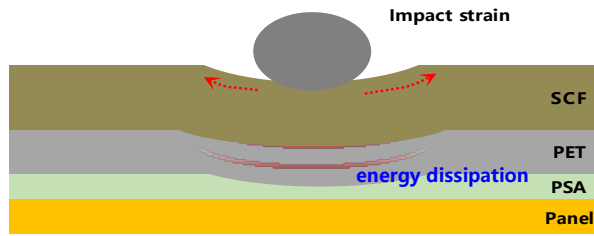


Figure 4. Drop impact energy dissipation

2. PSA modulus and drop ball impact verification

2.1 Wettability (Attachment modulus) and ball dropping ability

The previous text has already explained the relationship between modulus and wettability through JKL contact theory. Now, we are developing Acrylic UV PSA adhesives with different moduli to characterize the relationship between wettability and product anti ball ability; The T_g point of this adhesive system can be designed to be below -20°C . Prior to UV, the internal cross-linking degree of the colloid is low, the movement space of the chain segments is large, the modulus at room temperature is $\leq 50\text{KPa}$, and the softness of the colloid is high. After UV, secondary cross-linking is carried out by exciting free groups and doping reactants, and the cross-linking density is increased. The modulus at room temperature can be increased to the range of $100\text{-}500\text{KPa}$, and the colloid hardness is improved.

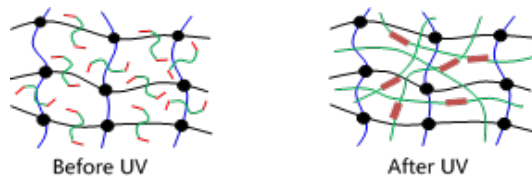


Figure4. Crosslinking Density of Acrylic PSA

To visually demonstrate the impact of wettability on drop-ball performance and ensure it remains the sole variable, the same Acrylic UV PSA adhesive was used under two different processing flows:

Process Flow 1: The first sample was laminated onto the panel surface before UV curing, during which its modulus was $\leq 50\text{ kPa}$. It was then UV-cured to increase the modulus to over 100 kPa .

Process Flow 2: The second sample was first UV-cured to achieve a modulus of $>100\text{ kPa}$ before being laminated onto the panel surface.

After sample preparation, a test model was established. It is understood that the primary cause of panel failure is excessive peak impact force, which leads to fractures in the internal film layers. By measuring the difference in the impact force transmitted to the underside of the PI layer, the buffering effectiveness of the PSA adhesive combined with the PI layer under different bonding conditions can be qualitatively evaluated.

Refer to the test model in Figure 5. The buffer rate is calculated as follows:

$$\text{Buffer Rate} = (\text{No-load Impact} - \text{Sample Buffering Impact}) / \text{No-load Impact} \times 100\%$$

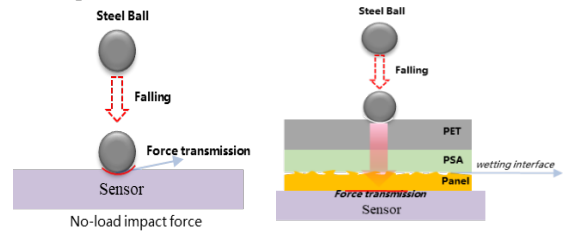


Figure 5. Test Model for Energy Buffer Rate

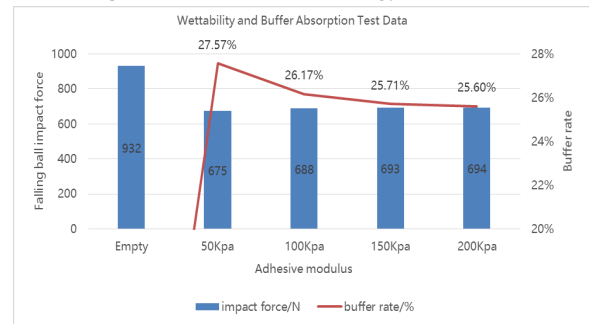


Figure 6. modulus and buffering rate

Through the verification of the above data, it further confirms that the JKL contact theory can be applied to Acrylic PSA adhesive, which synchronously indicates that the lower the modulus during attachment, the better the wettability, and the sufficient contact with the pasted interface. During impact, stress concentration points are avoided, weakening the peak impact force transmitted to the interior of the panel and improving the drop ball impact performance of the screen to a certain extent.

2.2 Low-Temperature Modulus and Drop Ball Performance Verification

In addition to the improved drop impact strength achieved through the high wettability of the Acrylic PSA, the modulus of the adhesive itself at the moment of impact also significantly influences the peak impact force and energy dissipation efficiency. If the drop impact event is regarded as a dynamic system with a millisecond-level response during the instant of impact, it is essential to focus on analyzing the effect of high-frequency modulus on drop impact performance.

a) Mechanical Analysis Using the Momentum Theorem

$$F \cdot \Delta t = m \cdot \Delta V \quad (4)$$

The change in kinetic energy of the ball is equal to the impulse of the external force F , as shown in Figure 4. This force F corresponds entirely to the peak impact force exerted on the screen body during the ball impact. Therefore, for the same drop height (i.e., constant momentum), a longer impact duration results in a lower peak impact force. To achieve this, it is necessary to increase the deformation of the PSA adhesive, thereby extending the impact time and reducing the peak impact force. This requirement corresponds to a lower modulus of the adhesive at high frequencies, which enhances its deformation capability.

b) From the perspective of energy dissipation, with reference to John Ferry's high-frequency energy dissipation theorem:

$$W_d = \pi \gamma \omega^2 G'' \quad (\gamma_0 \text{ is the strain amplitude}) \quad (3)$$

Refer to Formula 3., a higher value of G'' indicates a greater ability of the material to convert mechanical energy into heat per cycle. Especially, the dissipated energy is extremely sensitive to the deformation amplitude γ_0^2 (strain = compression displacement / sample thickness). If the strain amplitude doubles, the dissipated energy increases by a factor of four, demonstrating that deformation is the key driver of energy dissipation.

2.2.1 Temperature-Modulus Conversion

The drop impact test waveform obtained from the testing equipment is shown below. The impact pulse width is <0.15 ms, indicating that the contact time from the initial impact to the peak force is <0.075 ms.

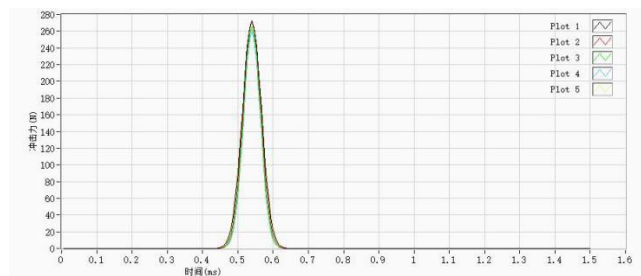


Figure 7. Falling ball impact waveform diagram

Based on the contact time of <0.075 ms, the impact frequency f exceeds 1 GHz. This frequency lies beyond the measurement range of conventional rheometers. By applying the time-temperature superposition (TTS) principle, the high-frequency modulus at room temperature can be characterized using low-frequency data obtained at lower temperatures, with reference to the Williams-Landel-Ferry (WLF) equation.

$$\log aT = \frac{-C_1 (T - T_{ref})}{C_2 + (T - T_{ref})} \quad (5)$$

For the PSA adhesive, the WLF constants are $C_1 = 17.44$ and $C_2 = 51.66$, $T_{ref} = 25^\circ\text{C} = 298.15\text{ K}$. The principle of modulus equivalence: $G(T, 1\text{Hz}) = G(25^\circ\text{C}, >1\text{GHz})$; According to the time-temperature superposition principle, the shift factor aT satisfies Equation

$$aT = \frac{f_{at} \cdot T_{ref}}{f_{at} \cdot T} \quad (5)$$

Therefore, the shift factor aT is negatively correlated with the shifted temperature T . That is, the higher the instantaneous impact frequency, the lower the corresponding shifted temperature T . Consequently, the modulus of the Acrylic PSA is lower at this lower temperature, making it more prone to strain buffering and energy dissipation upon impact.

Two acrylic UV PSA adhesives, denoted as Sample A and Sample B, were prepared with significant differences in modulus at low temperatures. The rheological test results of modulus versus temperature at 1Hz frequency are as follows. Below 0°C , the modulus of Sample A consistently remains lower than that of Sample B, meaning that under high-frequency impact conditions equivalent to small ball impacts, the modulus of Sample A is always lower than that of Sample B. This ensures that Sample A generates greater strain and dissipates impact energy when subjected to impacts, as further explained in the subsequent simulation

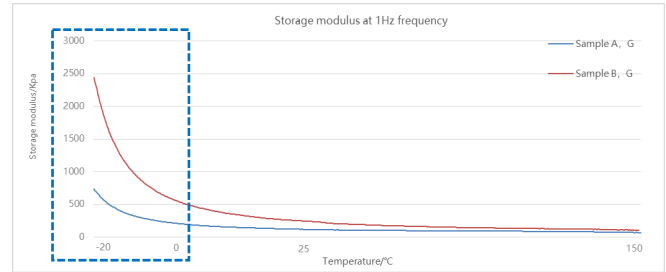


Figure 8. Modulus vs. temperature curve at 1 Hz

The two types of PSA were applied in the same OLED panel module structure, with BP-PSA as the only variable factor. A ball drop impact test was conducted, and the failure height data (displaying bright spots and dark spots) of the panel under ball drop impact were recorded for reference in Figure 9.

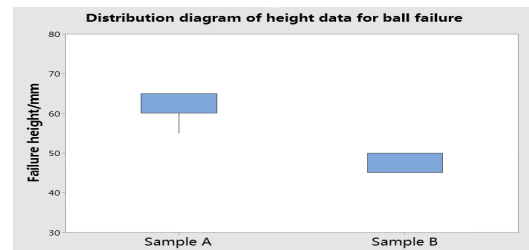


Figure 9. Falling ball failure height data

Based on the drop ball failure height data in Figure 9, the drop ball resistance range of panels using Sample A adhesive is 55–70 mm, while that of panels using Sample B adhesive is 40–50 mm.

The ability to increase the intensity of falling balls is about 20% or more.

3. Simulation Data Extension

Based on dynamic mechanics, contact mechanics, and viscoelastic theory, Establish a drop-ball impact simulation platform.. The simulation structure refers to Figure 10. By inputting the frequency-modulus curves and stress-strain curves of BP-PSA adhesive, the contour evolution of impact strain from the surface to the internal structure is presented as follows. It can be clearly seen that the strain of Sample A PSA adhesive with low modulus is higher than that of Sample B PSA adhesive.

Select the panel layer located directly under the PSA adhesive, adjust the upper and lower limits of strain, update the strain cloud map of this layer, and obtain the maximum strain of the panel layer under Sample A PSA, which is $2.825E^{-3}$. Referring to Figure 11, the maximum strain of the panel layer under Sample B PSA is $4.106E^{-3}$. Referring to Figure 12, By comparing the data, it can be clearly observed that the strain level of the panel layer below Sample A PSA has decreased by 67%, which proves that the PSA adhesive with lower modulus has better energy buffering at low temperatures, reduces the energy transmitted to the panel, lowers the strain of the panel, and improves the impact resistance of falling balls

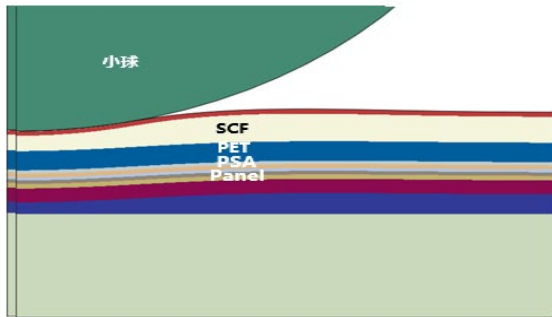


Figure10. Simulation model

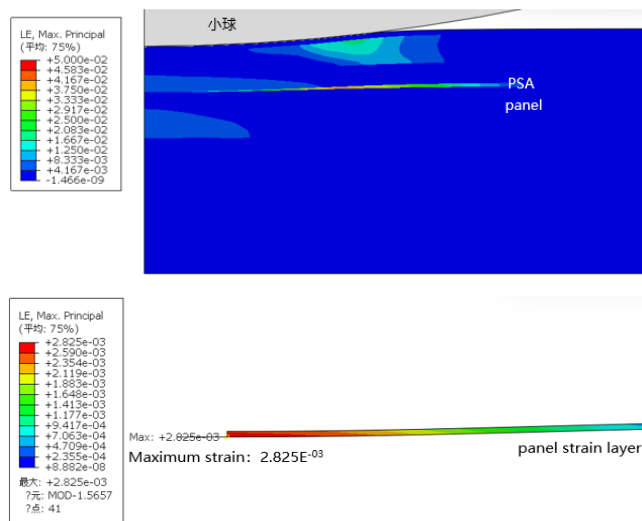


Figure11. PSA Sample A strain cloud map

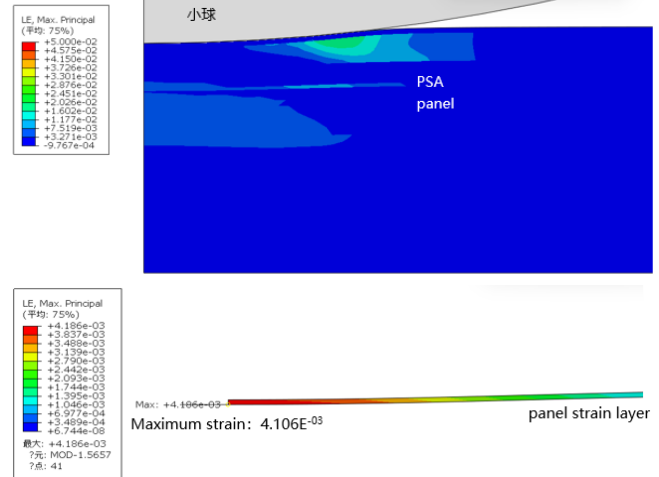


Figure12. PSA Sample B strain cloud map

4. Conclusion

In summary, both JKL contact theory and John Ferry energy dissipation mechanism are applicable for analyzing the strength performance improvement of flexible OLED displays under drop impact; When the PSA is being attached, the modulus (25 °C, 1Hz) is reduced from 200Kpa to 50Kpa, which improves the wettability and increases the contact area of the adhesive, The drop Impact strength can be increased by 2%. By controlling the low-temperature modulus (-20 °C, 1Hz) within 1000Hz and increasing the strain of the PSA adhesive during high-frequency impact, the energy dissipation effect of the adhesive is improved, and the drop Impact strength is increased by more than 20%. Based on this, the existing analysis results can provide some inspiration and guidance for the trend of thin and lightweight technology in the flexible OLED display industry, while improving the impact strength of Flexible OLED Panels

5. References

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